What Does 99% F1 Mean on ISOT?

CS 1675 Project | Milestone 2 | 25 October 2026 | github.com/Dande-lifeon/fake-news-detection

Group members: 王溪筠 (2024141520230), 何水 (2024141520241), 王浩懿 (2024141520231)

Data

All counts below come from pandas. ISOT contains 44,898 articles: 21,417 Reuters articles (real, label 0) and 23,481 from flagged sites (fake, label 1). We drop 844 bodies shorter than 50 characters (631 of them empty), then 5,551 exact duplicates (zero cross-label conflicts), leaving 38,503 (21,194 real / 17,309 fake). Cleaning removes 26.3% of fake but only 1.0% of real rows, shifting the fake share from 52.3% to 45.0%. The subject column is disjoint across classes, and 55% of real versus 9% of fake articles fall in Sep-Dec 2017; both fields are audit-only. The stratified 70/10/20 split gives 26,951 / 3,851 / 7,701; the test split is locked and read once per condition.

Protocol

TF-IDF is fit on train only. C is chosen by 5-fold cross-validation on train; alpha is chosen on val. Models are refit on train+val. For a common operating point, Table 1 reports test F1 at threshold 0.5; validation-tuned thresholds are saved as a sensitivity check. The positive class is fake. We report precision, recall, F1, ROC-AUC, PR-AUC, confusion counts, 95% bootstrap CIs (1,000 stratified resamples), and McNemar tests for paired comparisons. No test result feeds back into tuning.

Models

Each article has a title and a body. We compare four conditions: title-only, body-only, early concat, and late fusion (alpha p_title + (1-alpha) p_body). Logistic regression (LR) and a calibrated linear SVM use the same splits; alpha is 0.40 for LR and 0.50 for SVM. We also train an 18-feature format-only LR on surface cues from title and body - dateline, URL, Getty/featured-image credits, casing, punctuation and length - with no body vocabulary. The final probe is a single rule: no (Reuters) means fake.

Results

Table 1. Test F1 (fake class, threshold 0.5, n = 7,701).

Model	title	body	concat	fusion	alpha
LR	0.9429	0.9929	0.9962	0.9962	0.40
SVM	0.9425	0.9957	0.9975	0.9970	0.50

Test F1 for the fake class. Body dominates title. Fusion and concat are statistically indistinguishable (LR p=0.80; SVM p=0.39), and LR versus SVM is also within noise. Fusion F1 is 0.9962 [0.9947, 0.9975] for LR and 0.9970 [0.9957, 0.9981] for SVM; the intervals overlap.

The Shortcut

A single (Reuters) rule reaches F1 0.9943; the 18-feature format model reaches 0.9953; full text reaches 0.9962 (LR) and 0.9970 (SVM). In the test split, 99.08% of real bodies contain (Reuters), against 0.03% of fake. Masking dateline, URL and media credits and retraining lowers fusion only to 0.9928/0.9903 (LR) and 0.9944/0.9913 (SVM). Title-only is unchanged. The shortcut is not one cue; it is distributed across title style, topic and residual source words.

Problems

First, single-cue masking is insufficient: residual F1 stays above 0.99, so a clean content-only model is hard to identify. Second, the test set has only 21-26 fusion errors. Fusion and concat differ by at most 0.0007 F1, LR versus SVM is 0.0007, and the 18-feature model versus full text is 0.0009-0.0017; the confidence intervals overlap. The benchmark has no statistical room to rank models. Third, labels are source labels, 1.3% of test rows share their first 200 characters with a training row, and eight planned conditions (2 models x 4 conditions) have been evaluated on test. We can claim source-style recognition, not fact-checking.

Next Plan

Next we will (i) build a stronger cue ladder and a cue-balanced or temporal test, (ii) run Bi-LSTM and DistilBERT under the same original/masked protocol and compare delta F1, and (iii) audit 100-200 labels. The goal is not a higher score; it is a defensible measurement of what the score contains. Code, metrics, CIs and figures are in the repository. The pipeline, baselines, shortcut and masking study are complete; neural code is ready but not trained.

Data: ISOT Fake and Real News, Kaggle (clmentbisaillon/fake-and-real-news-dataset), ISOT Lab, University of Victoria; CC BY-NC-SA 4.0. Source study: Ahmed, Traore & Saad, ISDDC 2017, doi:10.1007/978-3-319-69155-8_9.